

CSA0711
SIMATS ENGINEERING
ASSESSMENT TOOL 3

9

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I Dinesh Bhavadas (192572028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Bhavadas

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 marks
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	2/1
Optimization	20%	2	2	2	2	2/2
Analysis	20%	1	1	1	1	1/1
Code Quality	15%	1	1	1	1	1/1
		6	6	6	4	4

26/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Debugging and Optimization of IPv4

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (126)

Default Gateway: 192.168.10.1

Identify the subnet:-

Subnet mask 126 means:

Block size = 64

Subnets are:-

1. 192.168.10.0/26
2. 192.168.10.64/26
3. 192.168.10.128/26
4. 192.168.10.192/26

The IP address 192.168.10.65 belongs to : 192.168.10.64/26

Network details

Network address: 192.168.10.64

Broadcast address: 192.168.10.127

valid host range: 192.168.10.65 to 192.168.10.126.

Verify IP and gateway:-

IP address

192.168.10.65 is valid.

Gateway:-

Gateway = 192.168.10.1

192.168.10.1 belongs to subnet:

Network: 192.168.10.0/26

Configuration error:-

The workstation belongs 192.168.10.64

but the gateway belongs to 192.168.

optimized configuration:-

option-1:

IP address: 192.168.10.10

Gateway: 192.168.10.1

option-2:

IP address: 192.168.10.65

Gateway: 192.168.10.66

usable hosts: 62.

Debugging IPv6 Configuration

Device A:-

2001:db8:ffff:00:42:8329:164

Expand IPv6 addresses:-